

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First / Second Semester of B. Tech. Examination (CE/IT/EE/EC/ME/CL)

December 2012

CE101 – Fundamentals of Computing & Programming

Date: 08.12.2011, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q - 1 (a) Do as directed.

1. Do conversion with proper calculation. [02]

(i) $(1101\ 0101\ 0110)_2 = (?)_8$ (ii) $(420)_{16} = (?)_{10}$

2. Draw the flowchart to add 1 to N numbers. [02]

3. Classify valid and invalid variable name in C. [03]

(i) ASCII

(ii) int_main

(iii) Two2

(iv) size_of

(v) char_getch

(vi) 1A

(b) What is the correct output of the following code? [04]

(1) main()

```
{  
    int a=-3, b=-5, c=-2;  
    int z=(a>b)?((a>c)?a:c):((b>c)?b:c);  
    printf("%d", z);  
}
```

(2) main() {

```
    int x,y,z =3%3;  
    for(x=1,y=2;y<8;x++)  
    { printf("%d\n", z);  
      y=y+2;z++;  
    }  
}
```

- Q - 2 (a) Write a C program to add all the Even numbers from N1 to N2, where N1 and N2 will be user dependent. [05]

- (b) Write a C program to find the maximum number from the array of 10 floats. [05]

- (c) Evaluate the following expressions step by step according to C compiler. [02]

$3 + 2 / 3 \% 2 * 3 - 2 * 3 + 2$

OR

- Q - 2 (a) Write a C program to evaluate the series: $2^3 + 4^3 + 6^3 + \dots + N^3$, where N will be entered from the user. [05]

- (b) Write a C program to reverse the entered string without using *string.h*. [05]

- (c) Write the equivalent code using while statement. [02]

```
int i, j, k;  
for(i=9, j=6; j<6; j=j+1/2)  
{ k=i++ - --j;  
  i=++i + i++;  
  printf("%d %d", i, j); }
```

Q - 3 Answer the following questions. [12]

1. What are the advantages to use user-defined functions in a program?
2. Explain any four built-in functions related to string.
3. Differentiate: Structure vs. Union

OR

Q - 3 Answer the following questions.

[12]

1. Explain recursion with example.
2. Which are the modes supported by *fopen()* for file? What are the reasons to close a file using *fclose()*?
3. Differentiate: Call by value vs. Call by address

SECTION – II

Q - 4 (a) Answer the following.

[07]

1. Differentiate: Compiler vs. Interpreter
2. Which symbols are used to indicate *end of string*, *tab* and *new line*?
3. Explain C tokens.
4. What is the use of indirection operator in pointer?
5. Show at least two methods to initialize the string.
6. Specify life of a local, static and global variable.
7. `int a[5]; a[5]; a[0]=5;` Explain the role of 5 in each statement.

Q - 4 (b) Mention the following statements are True or False. If false provide reason.

[04]

1. `#define` works as a global variable in any program.
2. `(type)` is a unary operator in C.
3. C compiler does not support bound checking for any type of array.
4. If `p1` and `p2` are two integer pointers, `p1 + p2` is valid statement.

Q - 5 (a) Write programs in C.

[10]

1. Write a program to check whether the entered number is prime number or not.
2. Write a program to input, exchange and print value of two variables using pointers.

(b) Write a C program to find whether entered number is odd, even or zero.

[02]

OR

Q - 5 (a) Write programs in C.

[10]

1. A structure has data members: `cm` and `mm`. Write a program that enter two variables, add them and assign into third variable. Validate and print the answer.
(e.g. 16mm = 1 cm and 6 mm)

2. Write a program to find the sum of individual digits of a given integer.

(b) Write a program to print 100 to 1 numbers, where N is user dependent.

[02]

Q - 6 (a) Explain the unary, binary and ternary operators with example.

[04]

(b) Answer the following questions.

[08]

1. Explain the functions `isalpha()` and `toupper()`.
2. Explain the syntax of `switch()`.
3. What is pointer? How it can be initialized?
4. Write a short note on a global variable.

OR

Q - 6 (a) Draw the memory layout for each element of the given array: `int z[3][4]`. Base address is 1001.

[04]

(b) Answer the following questions.

[08]

1. Explain format specification to print an integer number.
2. Explain the nested if with example.
3. Write a short note on implicit type conversion.
4. Explain short hand operator and pre/post increment operator.
